

INDUSTRIAL TECHNOLOGY & APPLIED ENGINEERING

Official Ministry of Education Standard Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Official Egyptian Ministry textbook for Industrial Technology & Applied Engineering covering metallurgy, tensile testing, machining forces, induction machines, power factor, sensors, and OSHA safety standards.

CORE CURRICULUM HIGHLIGHTS:

- Hooke's law, ASTM tensile testing, and steel heat treatment
- Cutting speed calculations, lathe turning, and Chvorinov's casting rule
- Three-phase induction motors, Star-Delta starting, and power factor correction
- 4-20 mA current loops, RTD probes, and OSHA Lockout/Tagout safety protocols

APPROVED CHAPTER MODULES:

- Engineering Materials, Metallurgy & Mechanical Testing	pp. 1 - 64 (4 core units)
- Machining Technology, Metal Forming & Welding	pp. 65 - 128 (4 core units)
- Electrical Machines, Power Distribution & Industrial Safety	pp. 129 - 192 (4 core units)
- Industrial Electronics, Sensors & Applied Control Circuits	pp. 193 - 256 (4 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Industrial Engineering Formulas & Standards Compendium (8p - 24.2 KB)
Full Ministerial Textbook: 256 Pages - 26.8 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-IND-G12
Legal Deposit: 2025/14892 - ISBN: 978-977-12-8777-10



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Official Egyptian Ministry textbook for Industrial Technology & Applied Engineering covering metallurgy, tensile testing, machining forces, induction machines, power factor, sensors, and OSHA safety standards. Hooke's law, ASTM tensile testing, and steel heat treatment Cutting speed calculations, lathe turning, and Chvorinov's casting rule Three-phase induction motors, Star-Delta starting, and power factor correction 4-20 mA current loops, RTD probes, and OSHA Lockout/Tagout safety protocols

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
---	---	--

3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Engineering Materials, Metallurgy & Mechanical Testing	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Machining Technology, Metal Forming & Welding	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Electrical Machines, Power Distribution & Industrial Safety	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Industrial Electronics, Sensors & Applied Control Circuits	7 wks	15 pts	25.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: ENGINEERING MATERIALS, METALLURGY & MECHANICAL TESTING

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Engineering Materials, Metallurgy & Mechanical Testing.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Engineering Materials, Metallurgy & Mechanical Testing.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: MACHINING TECHNOLOGY, METAL FORMING & WELDING

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Machining Technology, Metal Forming & Welding.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Machining Technology, Metal Forming & Welding.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: ELECTRICAL MACHINES, POWER DISTRIBUTION & INDUSTRIAL SAFETY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Electrical Machines, Power Distribution & Industrial Safety.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Electrical Machines, Power Distribution & Industrial Safety.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

INDUSTRIAL TECHNOLOGY & APPLIED ENGINEERING - CORE FORMULAS	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{net} = 0, \sum M_{pivot} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$
Normalized Standard Score:	$Z = \frac{(X - \mu)}{\sigma}$
PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha \cdot T(u) + \beta \cdot T(v)$
Harmonic Synthesis Mean:	$H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary:	$s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = (\text{Output}_{actual} / \text{Input}_{standard}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: TH-IND-G12
Curriculum Cycle: 2025/2026 Academic Year | National Edition
Official Portal: <https://moe.gov.eg/ar/elearning-content/>
All rights reserved (c) 2025 Ministry of Education & Technical Education - Egypt.
Unauthorized commercial reprinting or distribution without ministerial sanction is prohibited.